Colive API Architecture – QA/LIVE Switching

Colive website rebuild (Next.js, `website-react`) · Branch `Dev-Hari` · Work date: 3 Nov 2026 · Status: uncommitted

> All calls to the Colive backend now go through one central layer. A single setting, `COLIVE_ENV` in `.env.local`, decides whether the site uses the QA or the LIVE APIs. Endpoints are listed once in a registry, browser calls go through one proxy route, and adding a new API means adding one registry entry and one typed function.
>
> Two follow-up issues were fixed: a 404 on `GetFilterMasterDetails` (incorrect QA base URL) and a page-level error when scrolling search results (image host not allowed).

1. What changed

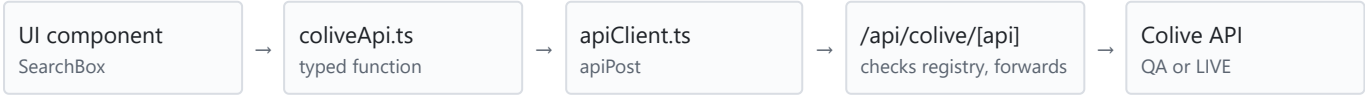
Before this change, two separate clients called the backend. The server-side search and factsheet code called the CRM API at a fixed live URL. The home-page search box called the website API through a rewrite in `next.config.ts`. There was no way to switch to QA.

File	Status	Purpose
`src/lib/server/apiConfig.ts`	New	Reads `COLIVE_ENV` and returns the QA or LIVE base URL.
`src/lib/server/coliveEndpoints.ts`	New	Endpoint registry: method, host, and one path or separate QA/LIVE paths for each endpoint.
`src/lib/server/coliveFetch.ts`	New	Server helper that calls a registered endpoint in the active environment.
`src/lib/api/apiClient.ts`	New	`apiGet` / `apiPost` for browser code; shared response handling.
`src/app/api/colive/[api]/route.ts`	New	Single proxy route for browser calls; forwards registered endpoints only.
`.env.example`	New	Documents the environment settings; can now be committed.
`src/lib/coliveApi.ts`	Changed	Typed browser functions (suggestions, place lookup) now use `apiPost`.
`src/lib/propertySearch.ts`, `propertyDetails.ts`	Changed	Use the registry through `colivePost` instead of a hard-coded live URL.
`next.config.ts`	Changed	Old `/api/Colive` rewrite removed; second image location allowed (section 6).
`.gitignore`	Changed	Exception added for `.env.example`.

The existing routes `/api/property-highlights/[link]` and `/api/visit-slots` are unchanged.

2. How a request flows

Browser components (e.g. home-page search box)



Server-rendered pages (search, property factsheet)

Page / server component	→	propertySearch.ts / propertyDetails.ts	→	coliveFetch.ts colivePost	→	Colive API QA or LIVE

Server pages call the backend directly instead of going through the app's own proxy route. This avoids an extra network round-trip, and they still use the same registry and environment setting.

3. Environment configuration

```
# .env.local
COLIVE_ENV=LIVE                                    # QA or LIVE
COLIVE_QA_API_URL=https://api.colive.com/web/qa/    # base URL only
COLIVE_LIVE_API_URL=https://api.colive.com/web/live/  # optional; this is the default
COLIVE_LIVE_WEB_URL=https://www.colive.com/          # optional; this is the default
```

- Base URLs must stop at `/web/qa/` or `/web/live/`. The code adds the endpoint path, e.g. `CRMapi/PropertyCRM/GetFilterMasterDetails`.
- Only the selected environment is called. If the QA URL is missing while QA is selected, the app fails with an error naming the missing setting, so it never falls back to LIVE without telling anyone.
- Next.js reads `.env.local` only at startup, so restart the dev server after changing it.

4. Registered endpoints

Registry name	Host	QA path	LIVE path	Checked
GetPropertyDetails (search)	CRM	`CRMapi/PropertyCRM/GetPropertyDetailsV3Web`	`CRMapi/PropertyCRM/GetPropertyDetails_V3_Web`	QA + LIVE
GetFilterMasterDetails	CRM	`CRMapi/PropertyCRM/GetFilterMasterDetails` (same for both)		QA + LIVE
GetPropertyDetailsByIdForWeb_V2	CRM	`CRMapi/PropertyCRM/GetPropertyDetailsByIdForWeb_V2`		LIVE
GetAvailableOccupancyType..._Factsheet	CRM	`CRMapi/PropertyCRM/GetAvailableOccupancyTypeByPropertyId_Factsheet`		LIVE
GetScheduleVisitDays	CRM	`CRMapi/PropertyCRM/GetScheduleVisitDays`		LIVE
GetSuggestionsBasedonSearchText	Website	`api/Colive/GetSuggestionsBasedonSearchText`		LIVE
GetDetailsByPlaceId	Website	`api/Colive/GetDetailsByPlaceId`		LIVE

CRM host: `api.colive.com/web/qa/` or `/web/live/`. Website host: `www.colive.com`, which only accepts requests that come from its own site, so the proxy sends that origin.

5. Differences from the proposed design

Proposed	Implemented	Reason
One base URL per environment	Each endpoint also names its host (CRM or website)	The suggestion and place-lookup APIs do not exist on the CRM API; they are only on `www.colive.com/api/Colive/`.

All calls go through `/api/colive/[api]`	Browser calls only; server pages call the backend directly	Avoids the server calling itself over HTTP; same registry and environment setting.
GetReelsDetails, GetEvents registered	Not registered	The `Community` controller does not exist on the live API, and these endpoints are not used yet.

6. Issues found and fixed

Issue	Cause	Resolution
Search page error: "GetFilterMasterDetails failed: 404"	The QA base URL in `.env.local` was `https://api.colive.com/`, missing `/web/qa/`.	Switched to `COLIVE_ENV=LIVE` for now. Error messages now include the full request URL so this kind of mistake is visible immediately.
"We couldn't load homes right now" after scrolling to the end of results	Six properties use photos under `www.colive.com/assets/images/`, which was not an allowed image location. The error appeared once those cards rendered.	Location added in `next.config.ts`. All card images across the six cities (852 image URLs checked) now fall in the two allowed locations.

7. Verification

- TypeScript check, ESLint and production build all pass.
- Against LIVE: city page, `/search`, property page, visit slots, place suggestions and place lookup all returned data.
- The proxy rejects unregistered names, wrong HTTP methods and built-in names such as `toString` with a 404.
- QA search and filter endpoints were checked directly and return data. The app itself has not yet been run against QA.

8. Open items

Item	Owner	Note
Set `COLIVE_QA_API_URL=https://api.colive.com/web/qa/` in `.env.local`, then test with `COLIVE_ENV=QA`	Development	Currently running on LIVE.
Confirm QA paths for the factsheet, visit-slot and occupancy endpoints	Development / API team	Assumed to match LIVE.
Correct paths for GetReelsDetails and GetEvents	API team	Add to the registry when needed.
QA host for the website API (suggestions, place lookup)	API team	QA uses the live website API until `COLIVE_QA_WEB_URL` is set.
Check image hosts used on property-page galleries	Development	Only search-card images were checked.
Commit the changes	Development	All work is uncommitted on `Dev-Hari`.

9. Adding a new API

1. Add an entry to `src/lib/server/coliveEndpoints.ts` with its method, host and path (or separate QA and LIVE paths).

2. For browser use, add a typed function in `src/lib/coliveApi.ts` that calls `apiPost("Name", payload)`.
3. For server use, call `colivePost("Name", payload)` from a file in `src/lib`.

```
// coliveEndpoints.ts
GetEvents: { method: "POST", service: "crm", path: "CRMapi/<Controller>/GetEvents" },

// coliveApi.ts
export const getEvents = (payload: object) => apiPost<EventItem[]>("GetEvents", payload);
```